	Fewer Params	Precise Control	Support Phenomena
Finetune	✗	✗	+
Parameter-efficient Fine-Tuning	✔	✗	+
Knowledge Augmentation	○	✗	+
Continual Learning	✗	✗	+
Model Unlearning	○	✗	−
Knowledge Editing	✔	✔	+ −

Table 1: Integrated comparison between knowledge editing and related techniques. The symbol ✔ denotes the presence of a particular feature in the technique, while ✗ signifies its absence. + indicates an enhancement of the LLMs' capabilities, whereas − signifies a reduction or removal of certain abilities within the model.

Continual Learning Continual learning (CL), also known as lifelong machine learning or incremental learning, refers to the ability of machine learning models to continuously acquire new skills and learn new tasks while retaining previously learned knowledge [132–135]. This is akin to how humans learn throughout their lifetimes by continually accumulating new information and skills without forgetting the old ones. Conventional machine learning models struggle with this as they are trained on independent and identically distributed data. When the distribution shifts or new tasks are encountered, their performance significantly degrades on older tasks due to catastrophic forgetting. Some key techniques being explored include replay-based methods [136, 137], regularization-based approaches [138, 139], and dynamic architecture methods [140, 141]. Continual learning focuses on allowing machine learning models to learn new tasks and adapt to new domains over time without forgetting earlier ones, which resembles the goal of knowledge editing. In contrast, knowledge editing focuses specifically on manipulating and updating the internal knowledge representations learned by pre-trained language models without regard to the underlying tasks or domains. The goal of knowledge editing is to dynamically refine language understanding independent of eventual applications, addressing the "fixedness" issue of pre-trained language models once deployed. Both areas are important for developing AI systems that can progressively acquire and flexibly apply knowledge throughout their lifetime.

Machine Unlearning In addition, it is crucial for models to be capable of discarding undesirable (mis)behaviors, which aligns with the concept of machine unlearning [142–146]. Chen and Yang [147] proposes an efficient unlearning framework EUL that can efficiently update LLMs without having to retrain the whole model after data removals, by introducing lightweight unlearning layers learned with a selective teacher-student objective into the Transformers. However, knowledge editing goes beyond unlearning by actively refining or erasing a model's learned knowledge base. Both machine unlearning and knowledge editing play important roles in enhancing reliability, fairness and effectiveness for LLMs across different domains and applications.

To conclude, the traditional approach to leveraging pre-trained language models involves fine-tuning them with target-specific data. However, in the realm of LLMs, this fine-tuning process encounters significant challenges. These include the vast number of parameters, substantial time and memory requirements, risks of overfitting, and issues like catastrophic forgetting. To address these challenges, several techniques have been developed, as we discussed above. Among these, knowledge editing emerges as a notable strategy. As we discussed in Table 1, knowledge editing, intersecting with these techniques, draws inspiration from a range of methodologies, showing promising results. This approach distinctively targets the knowledge embedded within LLMs, leveraging the inherent knowledge mechanisms of these models. Unlike simple adaptations of existing methods, knowledge editing necessitates a deeper comprehension of how LLMs function. It is not just about applying known techniques to new models; it is about understanding and manipulating the nuanced knowledge storage and processing capabilities of LLMs. Furthermore, knowledge editing represents a more precise and granular form of model manipulation as it involves selectively altering or enhancing specific aspects of a model's knowledge base, rather than broadly retraining or fine-tuning the entire model. These characteristics make knowledge editing a potentially more efficient and effective way to update and optimize LLMs for specific tasks or applications.

3 Knowledge Editing for LLMs

3.1 Preliminary

The substantial training on diverse datasets has equipped LLMs with a wealth of factual and commonsense information, positioning these models as virtual knowledge stores [48, 148, 149]. This rich knowledge base has been effectively utilized in various downstream tasks, as evidenced by numerous studies [150]. Additionally, Wang et al. [151] have demonstrated the potential of LLMs in autonomously constructing high-quality knowledge graphs, bypassing the need for human supervision. Despite their promise, LLMs, in their current state as emerging knowledge bases, exhibit certain limitations. These deficiencies often manifest as inaccuracies or errors in their outputs during practical applications. An ideal knowledge base would not only store extensive information but also allow for efficient and targeted updates to rectify these errors and improve their accuracy. Recognizing this gap, our paper introduces the concept of **knowledge editing** for LLMs. This approach is designed to enable quick and precise modifications to the LLMs, allowing them to generate more accurate and relevant outputs. By implementing knowledge editing for LLMs, we aim to enhance the utility of LLMs, moving them closer to the ideal of becoming universally reliable and adaptable repositories of knowledge. This advancement promises to address the current shortcomings of LLMs and unlock their full potential as dynamic and accurate knowledge bases for applications.

3.2 Task Definition

The initial goal of knowledge editing is to modify the specific knowledge k in the LLM and improve the consistency and performance of the LLM without fine-tuning the whole model. This knowledge can be associated with many areas and types, such as facts [79], commonsense [152], sentiment [153] and so on. Knowledge editing is challenging due to the distributed and entangled nature of knowledge in LLMs.

Suppose the original model is θ and given the knowledge k to be changed, by knowledge editing process F, we would get the post-edited model $\theta^{'}$:

$$\theta' = F(\theta, k) \tag{3}$$

The post-edited model $\theta^{'}$ is supposed to override undesired model beliefs on the knowledge k and keep other knowledge intact:

$$\begin{cases} \theta^{'}(k) \neq \theta(k) \\ \forall k^{'} \neq k, \theta^{'}(k^{'}) = \theta(k^{'}) \end{cases} \tag{4}$$

As a knowledge base, it's paramount that knowledge editing cater to three fundamental settings: knowledge insertion, knowledge modification, and knowledge erasure.

Knowledge Insertion. As fields and entities progress, it becomes imperative for LLMs to assimilate emergent information. Knowledge insertion fulfills this by bestowing upon LLMs new knowledge previously outside their purview:

$$\theta' = F(\theta, \{\emptyset\} \rightarrow \{k\}) \tag{5}$$

Knowledge Modification. Knowledge modification refers to altering knowledge already stored in LLMs:

$$\theta' = F(\theta, \{k\} \rightarrow \{k'\}) \tag{6}$$

This can be classified into two categories:

- **Knowledge amendment** - This aims at rectifying the inaccuracies embedded in LLMs to ensure the delivery of accurate information. As vast repositories of knowledge, LLMs are prone to housing outdated or erroneous information. Knowledge amendment serves to correct these fallacies, ensuring that models always generate accurate, up-to-date information.
- **Knowledge disruption** - Modifying LLMs to answer counterfactual or error prompts. This is more challenging as counterfactual notions initially receive lower scores compared to factual knowledge, as shown by Meng et al. [79]. This necessitates more targeted modification efforts.

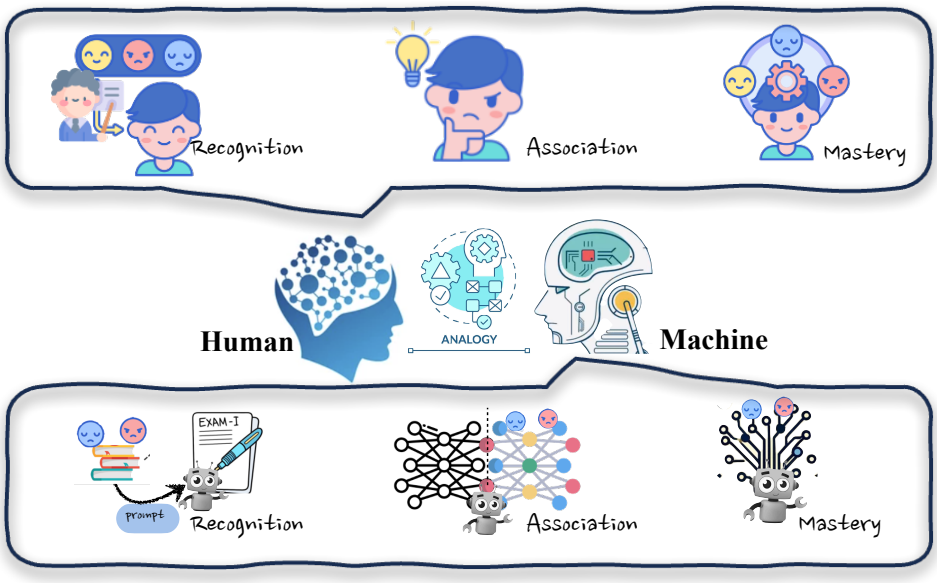


Figure 2: Applying Human Learning Phases [1–3] to Knowledge Editing in LLMs: We see an analogy of Human Learning Phases and Knowledge Editing in LLMs and categorize current knowledge editing methods based on the learning phases of humans: recognition, association, and mastery.

Knowledge Erasure. Knowledge erasure targets the excision or obliteration of pre-existing knowledge in a model, primarily to reset distinct facts, relationships, or attributes. Formally, we have:

$$\theta' = F(\theta, \{k\} \rightarrow \{\emptyset\}) \tag{7}$$

Implementing knowledge erasure is pivotal to expunge biases and noxious knowledge and to curtail the recollection of confidential or private data, thereby fostering responsible and trustworthy AI.

In conclusion, the interplay between knowledge insertion, modification, and erasure forms essential aspects of model editing techniques. When combined, these techniques empower LLMs to transform, self-correct, and ethically adapt as needed.

3.3 Methods

The development of LLMs has reached a point where their capabilities closely resemble human cognitive processes, especially in learning and acquiring knowledge. Drawing inspiration from how humans learn, we can analogously apply these concepts to the process of editing LLMs as Figure 2 shows. Educational and cognitive research [1–3] delineates human knowledge acquisition into three distinct phases: recognition, association, and mastery. These phases offer a framework for conceptualizing the methods of knowledge editing in LLMs[2] and we list them in Table 2.

- **Recognition Phase**: In the recognition phase, the model needs to be exposed to the new knowledge within a relevant context, just as people first encounter new information (§3.3.1). For example, providing sentences that illustrate a factual update as a demonstration of the model allows initial recognition of the knowledge to be edited.
- **Association Phase**: In the association stage, connections are formed between the new knowledge and existing knowledge in the model (§3.3.2), much like humans relate new ideas to prior concepts. Methods would combine or substitute the output or intermediate output $\boldsymbol{h}$ with a learned knowledge representation $\boldsymbol{h}_{\text{know}}$.
- **Mastery Phase**: The mastery phase involves the model fully acquiring the knowledge in their parameters and utilizing it reliably (§3.3.3), akin to deep human mastery. This method directly changed the model's weight, $\Delta \boldsymbol{W}$, and the model can deal with the problem without any external help or merge.

[2] https://github.com/zjunlp/KnowledgeEditingPapers.